INSIGHT: A Universal Neural Simulator Framework for Analog Circuits with Autoregressive Transformers

Souradip Poddar[1], Youngmin Oh[2], Yao Lai[1], Hanqing Zhu[1], Bosun Hwang[2], David Z. Pan[1]
[1]*ECE Department, The University of Texas at Austin, Austin, TX, USA,*
[2]*Samsung Advanced Institute of Technology, Republic of Korea*
souradippddr1@utexas.edu, youngmin0.oh@samsung.com, yao.lai@austin.utexas.edu,
hqzhu@utexas.edu, bosun.hwang@samsung.com, dpan@ece.utexas.edu

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***Abstract*—The compute-intensive nature of SPICE simulations hinders effective analog design automation. This paper introduces INSIGHT, a data-efficient, adaptive, high-fidelity, technology-agnostic universal neural simulator framework that formulates analog performance prediction as an autoregressive sequence generation task to accurately predict performance across diverse circuits. INSIGHT achieves test R^2 scores $\geq$0.95, outperforming existing neural surrogates. Cross-technology transfer learning experiments show that INSIGHT can preserve model performance with $\sim$60% less training data. Low-Rank Adaptation (LoRA) integration further reduces memory footprint by $\sim$42% and training time by $\sim$25%, maintaining high performance. Our experiments show that INSIGHT-based RL sizing framework achieves 100-1000$\times$ lower simulation costs over existing sizing methods for identical benchmarks and target specifications.**

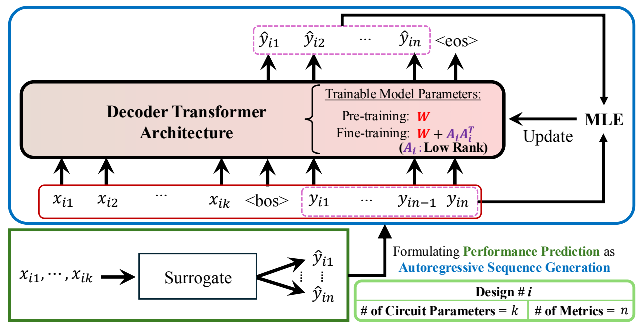


Fig. 1: INSIGHT Framework

I. Introduction

Analog circuits exhibit significant variation across and within types, each characterized by distinct performance metrics, design parameters of varying dimensions, and technologies. Consequently, this has spurred numerous optimization efforts to automate the labor-intensive analog design process. Bayesian Optimization methods [1], while known for their sample efficiency, often face scalability challenges in high-dimensional optimization for complicated analog circuit optimization tasks. On the other hand, learning-based approaches, like [2–10], require numerous real-time simulator interactions (i.e., poor sample efficiency). Given the vast and complex design space, the intricate interplay of underlying circuit physics in deep submicron technologies, and the multifaceted performance trade-offs, repeated simulations within the optimization loops become inherent for effective design space exploration for optimal circuit performance.

This heavy reliance on on-the-fly, time-consuming SPICE simulations motivates the development of cheap circuit simulator surrogates to expedite the analog optimization tasks. Traditional equation-based analog performance prediction methods [11, 12] are time-consuming, unscalable, and highly inaccurate in modern processes. Black-box ML-based performance modeling approaches employing polynomial regression [13], neural networks [14], and graph representations with neural networks [15] yield rapid inference but suboptimal prediction accuracies for complex circuits while still requiring substantial, costly simulation-based training data. [16] employs semi-supervised contrastive regression with scaling rules for synthetic data augmentation for effective analog performance prediction. While aiming for data-efficiency, this approach yields suboptimal R^2 scores with complex circuits on advanced processes across a broader range of performance metrics. Developing a *data-efficient*, *high-fidelity* circuit simulator surrogate with strong generalization capabilities and rapid fine-tuning potential remains an open challenge.

Given the high interdependence of performance metrics inherent to analog circuits, for the first time, we propose to formulate the analog circuit performance prediction problem as *an autoregressive generation task*, i.e., predicting simulation-costly performance specification by leveraging provided design parameters and partial performance information (provided or previously-generated) in a sequential manner. The autoregressive problem setup enforces the model to automatically reason and capture the inherent interdependency in the challenging analog performance prediction task, easing the learning complexity and providing much better prediction quality than previous learning-based methods [10].

In this paper, we introduce INSIGHT, a novel, technology-agnostic, data-efficient, high-fidelity universal neural circuit simulator framework that learns to accurately predict analog performance on previously unseen parameters autoregressively, using decoder-based transformer architecture [17, 18]. Our framework's autoregressive learning facilitates accurate prediction of simulation-costly critical transient specifications leveraging performance data from less expensive simulations. INSIGHT consistently demonstrates high R^2 scores and minimal test losses across various circuits and technologies, at microsecond inference times. Furthermore, INSIGHT is highly scalable and can integrate into existing EDA optimization workflows to enhance sample efficiency. INSIGHT supports data-efficient knowledge transfer across technology nodes and enables parameter-efficient fine-tuning on unseen nodes using Low-Rank Adaptation (LoRA) [19, 20], ensuring strong model performance and effective transfer learning.

The key contributions of this work include:

- **Novel autoregressive formulation**. We formulate analog circuit performance prediction as an autoregressive sequence modeling task, leading to our INSIGHT, a universal neural simulator framework designed around a decoder-only autoregressive Transformer architecture.
- **High data-efficiency with high prediction fidelity**: Our novel autoregressive formulation simplifies the learning complexity of the prediction task by inherently capturing the metric interdependencies and leveraging partial, less expensive metric information to boost prediction accuracy for the simulation-costly ones, thus enabling high-fidelity predictions with enhanced data efficiency.
- **A universal framework with circuit- and technology-agnostic capabilities**: The INSIGHT framework demonstrates strong generalization across unseen circuit types. Additionally, it supports efficient knowledge transfer to unseen technology nodes through parameter-efficient fine-tuning, ensuring high computational efficiency.
- We have rigorously validated INSIGHT across different circuits and technologies. Our cross-technology transfer learning experiments demonstrate INSIGHT's ability to maintain test R^2 scores $\geq$0.95 with $\sim$60% less training data than training from scratch. LoRA integration enhances computational efficiency, reducing memory footprint by $\sim$42% and training time by $\sim$25% while maintaining high model performance.
- To our knowledge, this marks the first **successful** application of a universal neural simulator framework leveraging autoregressive Transformers in analog design automation.

The source code[1] is available on GitHub.

II. Preliminaries

In this section, we briefly discuss some insights into analog circuit basics/metrics and Transformer architectures, discussing our rationale for choosing them.

A. Analog Circuit Basics and Metrics

In analog circuits, performance metrics and parameters are highly interdependent. By leveraging these interdependencies, we can build effective and data-efficient prediction models. For instance, in OPAMPs, fundamental metrics such as Quiescent Current (I_Q) and DC Gain, when combined with parameter information, can significantly enhance predictions for other key specifications, like unity-gain bandwidth (UGBW) and phase margin (PM). Similarly, integrating all static metrics provides crucial insights into the circuit's potential transient behavior. Thus, leveraging partial performance metric information can significantly enhance prediction accuracies for critical, simulation-costly specifications such as slew rate and settling time. Furthermore, given the transferable nature of circuit physics across technologies for a given topology, understanding performance interdependencies within one node provides valuable insights applicable to other nodes. These insights, combined with the adaptability and transferability of Transformer architectures, motivate the adoption of autoregressive Transformers to build a universal neural simulator framework that effectively predicts performance across technologies.

B. Transformer Architectures

Transformers constitute a powerful class of generative deep learning architectures that have revolutionized sequence modeling tasks in natural language processing and beyond [21–23]. These models stand out due to their self-attention mechanism, which effectively captures dependencies between tokens in a sequence regardless of distance. The self-attention mechanism [18] evaluates dependencies within a sequence by computing a weighted sum of value vectors. These weights highlight the most relevant token information for predicting subsequent tokens and are derived from the dot product of input vectors. Unlike traditional RNNs [24], Transformers process data in parallel, enhancing efficiency and scalability. While the original Transformer architecture [18] consists of stacked layers of encoders and decoders, decoder-only models like GPT [25] have recently demonstrated notable success in sequence data generation. Decoder-only Transformers employ solely the decoder component, composed of multi-head self-attention and position-wise feed-forward networks [26].

III. The INSIGHT Framework

In this section, we introduce our proposed universal neural simulator framework, INSIGHT, for analog performance prediction. Section III-A details INSIGHT architecture and the probabilistic rationale behind our approach. Section III-B discusses our performance metric sequencing strategy for data efficiency, while Section III-C explores INSIGHT's transfer learning capabilities.

A. INSIGHT Architecture

In INSIGHT, we are driven by a critical inquiry: Can we develop an *effective* and *efficient* surrogate model that utilizes circuit parameters and partial performance metrics information (if available) to predict remaining simulation-costly performance numbers accurately? This question is particularly pertinent as current analog front-end design automation heavily depends on time-intensive simulations like SPICE, incurring a strong demand to ease the reliance.

The problem can be formally represented as follows: For a given analog circuit topology and a B sequence of N design parameters $\{x_{i,1:N}\}_{i=1}^{B}$, where $x_{i,1:N} = (x_{i,1}, \cdots, x_{i,N})$, for example, width-to-length ratios (W/L), our goal is to predict the corresponding M performance specifications $\{y_{i,1:M}\}_{i=1}^{B}$ where $y_{i,1:M} = (y_{i,1}, \cdots, y_{i,M})$ accurately. In the context of maximum likelihood estimation (MLE), this predictive task is mathematically framed as maximizing the likelihood function $P(y_{i,1:M} \mid x_{i,1:N}; \theta)$, where θ represents the model parameters:

$$\begin{aligned}\hat{\theta} &= \arg\max_{\theta} \Pi_{i=1}^{B} P(y_{i,1:M}|x_{i,1:N}; \theta) \\ &= \arg\max_{\theta} \Pi_{i=1}^{B} \Pi_{j=1}^{M} P(y_{i,j}|x_{i,1:N}, y_{i,1:j-1}; \theta)\end{aligned}$$

[1] https://github.com/UTDA-group/INSIGHT-FRAMEWORK

This objective is equivalent to minimizing the Negative Log Likelihood (NLL), expressed as:

$$\mathrm{NLL} = -\sum_{i=1}^{B}\sum_{j=1}^{M} \log P(y_{i,j} \mid x_{1:N}, y_{i,1:j-1}; \theta) \tag{1}$$

Each term, $P(y_{i,j} \mid x_{1:N}, y_{i,1:j-1}; \theta)$, quantifies the likelihood of observing the token $y_{i,j}$, given the design parameters, the previously predicted performance specifications, and the model parameters. This formulation connects to the contextual approach in autoregressive sequence generation tasks involving decoder-based Transformers like GPT [25] and motivates us to use a decoder-style Transformer model for INSIGHT. Figure 1 presents a conceptual diagram illustrating INSIGHT's autoregressive sequential inferencing of performance metrics.

The formulation of circuit performance prediction as an autoregressive sequence generation task allows us to leverage the inherent strengths of transformer architectures. Unlike fully connected layers, which require fixed input and output dimensions, transformers' adaptive capabilities automatically accommodate varying lengths, which is crucial for circuits which have varying numbers of design parameters, performance metrics, and types. While recurrent neural networks (RNNs) can handle variable-length inputs, they suffer from inefficient training due to the vanishing gradient problem and limited parallel processing capabilities. Furthermore, the self-attention mechanism in transformers captures complex interdependencies among circuit parameters and performance metrics, potentially yielding more accurate predictions than standard deep neural networks or RNNs, especially for high-dimensional, heterogeneous circuit data. INSIGHT exploits these transformer strengths, facilitating the automatic construction of tailored high-fidelity performance prediction models within a unified framework that adapts to diverse parameter lengths and performance metric lengths. Note that INSIGHT's universality stems from its adaptable framework, which can consistently accommodate diverse circuit types and topologies across technologies and not a one-model-fits-all approach.

We implement INSIGHT using a custom decoder-only transformer architecture specifically tailored for analog circuit performance prediction with real-valued inputs and outputs. Unlike traditional language models, our approach skips tokenization and directly processes continuous values [21, 27]. This streamlined design results in a compact model with ~27.5K network parameters for a topology with 30 design variables and 20 performance metrics (total sequence length of 50). Compared to large language models (LLMs) with billions of parameters, INSIGHT's modest size enables computationally and data-efficient training from scratch, facilitating rapid training and inference. Table I presents INSIGHT's architecture details.

Remark 1: While the relationship between performance metrics, relative to design parameters and among themselves, is not inherently time-series, this formulation enables self-attention and autoregressive features to leverage their interdependencies for effective and efficient predictions.

TABLE I: INSIGHT Architecture

INSIGHT Parameter	Value
Network Architecture	Decoder-Based Transformer
Dim. of model	16
# of Heads	4
# of Decoder Layers	6
dim_feedforward	64
Activation	ReLU [28]
Learning rate	0.001
	with cosine annealing scheduler [29]

B. *Performance Metric Sequencing in INSIGHT*

In autoregressive sequence generation, accurate prediction of one metric significantly enhances the prediction accuracy of the subsequent, emphasizing the importance of metric sequence order for effective and data-efficient predictions. We employ a greedy strategy for metric sequencing in INSIGHT.

Algorithm 1 Metric Sequencing for Data-Efficient Prediction

Require: List of performance metrics $Y = [y_1, y_2, \ldots, y_M]$
Ensure: Ordered list of metrics $P_{ordered}$
  Sort Y in ascending order based on runtime and complexity
  $P_{ordered} \leftarrow []$ ▷ Initialize as an empty list
  while $|P_{ordered}| < |Y|$ **do**
    $y^* \leftarrow \arg\max_{y_i \in Y} \text{ExpectedInformationGain}(y_i | P_{ordered})$
    $P_{ordered}.\text{append}(y^*)$
    $Y.\text{remove}(y^*)$
  end while
  Sequence known dependent metrics after their dependencies in $P_{ordered}$
  return $P_{ordered}$

As outlined in Algorithm 1, metrics are initially sorted based on runtime complexity. Ties are resolved by iteratively selecting metrics that provide the highest expected information gain, given the previously predicted metrics, to prioritize data efficiency. If a specification is known to be directly derived from others, it is sequenced after its dependencies to optimize both data utilization and prediction accuracy. By deferring complex metrics that demand more data and exploiting performance interdependencies, this approach leverages simpler specifications to enhance the prediction accuracy of more complex ones, ensuring effective and data-efficient sequencing. Figure 2 illustrates the performance metric sequencing for the Folded Cascode Amplifier as an example, with arrows Left → Right indicating significant contributors to the autoregressive dependencies in the prediction model.

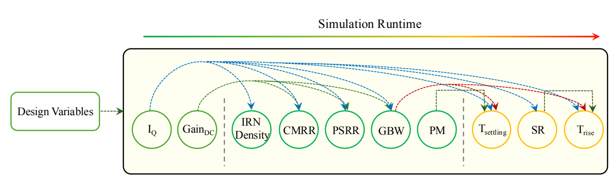


Fig. 2: Metric Sequencing in Folded Cascode Amplifier

Given a performance metric sequence order, INSIGHT utilizes self-attention to adaptively focus on the most relevant segments of the parameter (x_1 to x_N) and generated performance output (y_1 to y_{i-1}) sequence, to accurately predict each output metric y_i. Positional encoding is used to preserve sequence order and maintain context and coherence.

Our autoregressive performance prediction formulation, integrated with the INSIGHT framework, which comprises an adaptive architecture combined with our strategy for optimizing performance sequence order, offers a data-efficient and effective analog circuit performance prediction approach.

C. Transfer Learning with INSIGHT

The fundamental circuit physics, including performance interdependencies, trade-offs, and design principles inherent to a given topology, remain largely consistent and often translate across different process technologies. INSIGHT leverages transfer learning to facilitate data-efficient knowledge transfer between processes while maintaining high prediction accuracy. This same capability further enables rapid fine-tuning to new real data within the same process and topology, allowing swift model updates—a crucial feature for effective surrogate modeling in model-based optimization frameworks.

To further enhance computational efficiency, INSIGHT incorporates Low-Rank Adaptation (LoRA) [19, 20], as detailed in Algorithm 2, to reduce the model's memory footprint and fine-tuning time by updating only a small set of rank-decomposition matrices instead of all model parameters, while maintaining high prediction performance, as shown in Fig. 3.

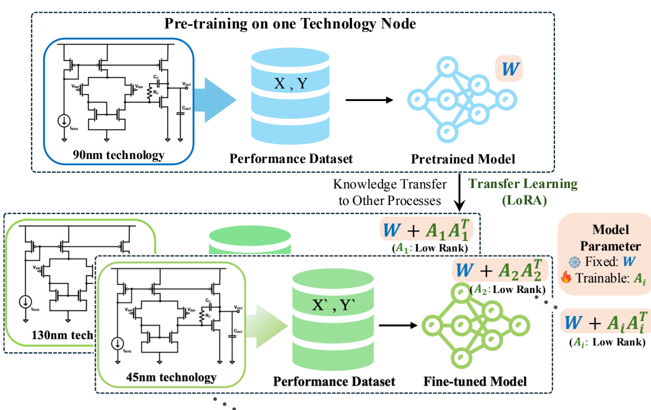


Fig. 3: LoRA-based Transfer Learning in INSIGHT

The overall wall-clock time for optimization in a model-based optimization framework can be approximated as:

$$\begin{aligned} T_{\text{optimization}} \approx t_{\text{overhead}} + K \cdot t_{\text{inference}} + N \cdot t_{\text{update}} \\ + M \cdot t_{\text{policy-opt}} + R \cdot t_{\text{real-interaction}} \end{aligned}, \tag{2}$$

where K, N, M, and R represent the number of surrogate model inferences, model updates, policy optimization steps, and real environment interactions during the optimization, respectively, with $t_{\text{inference}}$, t_{update}, $t_{\text{policy-opt}}$, and $t_{\text{real-interaction}}$ being their corresponding time durations, and t_{overhead} the time for data processing, one-time model pre-training, and other overhead tasks. INSIGHT aims to keep the critical components of Equation 2: N, R, t_{update}, and $t_{\text{inference}}$ small using its data-efficient high prediction accuracy, effective generalization, rapid fine-tuning capabilities, and fast ML-based inference, reducing the overall optimization time and enhancing the real-time sample efficiency.

Algorithm 2 LoRA-based Transfer Learning in INSIGHT

Require: Pre-trained $model$, $train\_dataset\_loader$, $device$, LoRA rank r, learning rate γ, epoch E, warm start step T_0
Ensure: Fine-tuned model with updated LoRA parameters
  function APPLY_LORA_TO_MODEL(network, LoRA_rank)
    for weight W in network **do**
      $W \leftarrow W.detach() + A_W A_W^T$ for a low rank learnable vector on A_W (LoRA parameter).
    end for
    return network
  end function
  $model$=APPLY_LORA_TO_MODEL($model$, r)
  $optimizer \leftarrow$ initialize Adam optimizer with LoRA parameters, learning rate γ
  $loss\_fn \leftarrow$ MSE loss function
  $num\_epochs \leftarrow E$
  $scheduler \leftarrow$ CosineAnnealingWarmRestarts with T_0
  for $epoch = 1$ to num_epochs **do**
    $total\_loss \leftarrow 0$
    $model.train()$
    for all $batch \in train\_dataset\_loader$ **do**
      compute output and loss
      backpropagate and update LoRA parameters
      aggregate loss
    end for
    update learning rate scheduler
  end for

IV. Experiments

A. Benchmark Circuits and Technologies

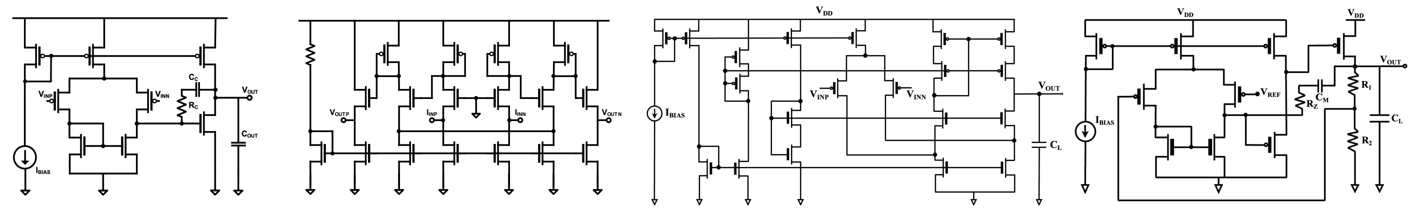
We trained and evaluated INSIGHT on a machine with eight NVIDIA P100 GPUs and an Intel Xeon E5-2698 V4 CPU. To demonstrate the effectiveness of INSIGHT, we compare its performance against two state-of-the-art neural surrogate approaches: Learn-by-Compare (LbC) [16] and the FC-Ensemble from CRONuS [10]. We use the following benchmark circuits:

I. 2-Stage Miller-Compensated OPAMP (NMOS Input Pair)
II. 2-Stage Miller-Compensated OPAMP (PMOS Input Pair)
III. 2-Stage TIA (Transimpedance Amplifier)
IV. 3-Stage TIA
V. Level Shifter

Additionally, we assess INSIGHT on the Folded Cascode Amplifier and Low-Dropout Regulator (LDO) using a commercial 45nm process with an extended set of performance metrics. Figure 4 provides detailed schematics for some of these circuits. Table II lists the performance metrics against which these circuits have been evaluated. We have validated our concept on 45nm, 90nm, and 130nm technologies.

B. INSIGHT Performance

1) Prediction Accuracy: An unseen test dataset of 1000 samples was held out for final evaluation. We created training sets of varying sizes; each split into an 80:20 train-validation



(a) (b) (c) (d)

Fig. 4: Schematics: (a) 2-Stage Miller-Compensated OPAMP (PMOS Input Pair), (b) 3-Stage TIA, (c) Folded Cascode Amplifier, (d) LDO

TABLE II: Performance Metrics for each circuit

<table>
<tr><td>Circuits</td><td colspan="10">Performance Metrics</td></tr>
<tr><td>Benchmark I</td><td rowspan="4">I_Q (mA)</td><td rowspan="4">DC Gain (dB)</td><td rowspan="4" colspan="2">Input-Referred Noise Density (IRN Density) @ 1KHz (V/$\sqrt{\text{Hz}}$)</td><td rowspan="4" colspan="3">CMRR (dB)</td><td rowspan="4">UGBW (MHz)</td><td rowspan="4" colspan="2">Phase Margin (°)</td></tr>
<tr><td>Benchmark II</td></tr>
<tr><td>Benchmark III</td></tr>
<tr><td>Benchmark IV</td></tr>
<tr><td>Benchmark V</td><td>DC Power (W)</td><td>Ratio</td><td colspan="5">Avg. Delay (sec)</td><td colspan="3">Delay Balance (sec)</td></tr>
<tr><td>Folded Cascode Amplifier</td><td>I_Q (mA)</td><td>DC Gain (dB)</td><td>IRN Density @1KHz (V/$\sqrt{\text{Hz}}$)</td><td>CMRR (dB)</td><td>PSRR (dB)</td><td>GBW (MHz)</td><td>Phase Margin (°)</td><td>Settling Time (sec)</td><td>Slew Rate (V/sec)</td><td>Rise Time (sec)</td></tr>
<tr><td>LDO</td><td>I_Q (mA)</td><td colspan="2">Phase Margin (°)</td><td>Efficiency</td><td>Dropout (mV)</td><td colspan="2">Load Regulation (%)</td><td>PSRR (dB)</td><td colspan="2">Line Regulation (%)</td></tr>
</table>

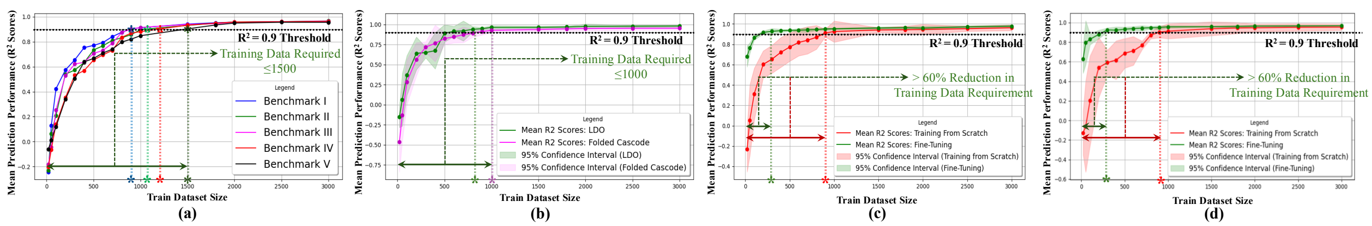


Fig. 5: INSIGHT Test Performance Plots: Demonstrating INSIGHT's high prediction accuracy and data efficiency

TABLE III: Test Performance Comparison Across Different Circuits: INSIGHT vs Baselines

	Test R^2 Scores for 2000:1000						
Neural Surrogate	**Benchmark I**	**Benchmark II**	**Benchmark III**	**Benchmark IV**	**Benchmark V**	**Folded Cascode**	**LDO**
FC Ensemble in CRONuS [10]	0.578316	0.493737	0.727980	0.627082	0.125414	0.317860	0.178161
LbC [16]	0.663117	0.540372	0.712673	0.695721	0.730163	0.521738	0.493361
INSIGHT	**0.957782**	**0.958691**	**0.957347**	**0.958614**	**0.950131**	**0.950587**	**0.976977**
	Test MSE Loss for 2000:1000						
Neural Surrogate	**Benchmark I**	**Benchmark II**	**Benchmark III**	**Benchmark IV**	**Benchmark V**	**Folded Cascode**	**LDO**
FC Ensemble in CRONuS [10]	0.02130	0.04518	0.04703	0.01640	0.04317	0.23961	0.17860
LbC [16]	0.01886	0.02964	0.03071	0.02377	0.02901	0.05928	0.06253
INSIGHT	0.00370	0.00793	0.00579	0.00427	0.00333	0.00360	0.00122

TABLE IV: Impact of Metric Sequencing on Data-Efficiency

Train Size	**Our Algorithm's Sequencing**	**Reverse Sequencing**
500	Mean R^2 = **0.824698**	Mean R^2 = 0.770055
1000	Mean R^2 = **0.926696**	Mean R^2 = 0.908772
2000	Mean R^2 = **0.950587**	Mean R^2 = 0.934486

TABLE V: Summary of Training Data Requirements

Benchmarks	**Training from Scratch**	**Transfer Learning**
Benchmark I	∼900	**∼300**
Benchmark II	∼1000	**∼200**
Benchmark III	∼900	**∼300**
Benchmark IV	∼1200	**∼300**
Benchmark V	∼1500	**∼400**

ratio. INSIGHT was trained on the training subset, validated for hyperparameter tuning and early stopping on the validation subset, and then evaluated on the fixed unseen test set. This process was repeated for 30 independent trials per training set size for statistical robustness, and the mean test perfor-

TABLE VI: Impact of LoRA on INSIGHT

Benchmarks	**# Trainable Parameters**		**Wall-clock Time Ratio**
	Full Fine-tuning (tune-all approach)	**LoRA**	($\frac{t_{\mathbf{LoRA}}}{t_{\mathbf{Full\ Fine\text{-}tuning}}} \times 100\%$)
Benchmark I	27.249K	**15.729K**	**∼72.21%**
Benchmark II	27.265K	**15.745K**	**∼71.13%**
Benchmark III	27.217K	**15.697K**	**∼74.22%**
Benchmark IV	27.313K	**15.793K**	**∼73.41%**
Benchmark V	27.233K	**15.713K**	**∼74.49%**

TABLE VII: Performance Comparison of INSIGHT-based RL framework (with PPO) vs CRONuS vs AutoCkt (PPO)

	INSIGHT-based framework (with PPO)	CRONuS [10]	AutoCkt (PPO) [6]
Benchmarks	Real-time Simulations	Real-time Simulations	Real-time Simulations
I	**2 ± 0.18**	450	14,100
II	**2 ± 0.18**	330	9,300
III	**15 ± 5**	990	11,100
IV	**15 ± 5**	390	8,700
V	**15 ± 5**	420	10,400

mances were recorded. Figure 5(a) showcases INSIGHT's mean test performance across benchmarks I–V when trained from scratch for varying training set sizes as described in the above setup for 90nm technology, while Figure 5(b) shows INSIGHT's performance on Folded Cascode and LDO on a commercial 45nm process. INSIGHT achieves high prediction performance ($R^2 > 0.9$) with $\leq$1500 training samples across benchmarks, highlighting its data efficiency. Table III summarizes INSIGHT's mean test performance using 2000 samples for training and validation following the setup described earlier. As shown, INSIGHT consistently achieves high test R^2 scores and low MSE losses across all benchmark circuits, outperforming our baselines [16] and [10]. Table IV highlights the effectiveness of INSIGHT's metric sequencing strategy for the Folded Cascode, showcasing improved data efficiency in limited data scenarios over the reversed order.

2) Knowledge Transfer across Technologies: For each topology, similar to the prediction accuracy evaluation setup discussed in Section IV-B1, we reserved an unseen test set of 1000 samples. INSIGHT's test performance was then evaluated on each for different training set sizes, comparing training from scratch against fine-tuning over 30 independent trials per training set size. Mean R^2 scores and standard deviations were noted. Figure 5(c) illustrates prediction performance on Benchmark II from 90nm to 130nm (to larger node), while Figure 5(d) shows results for Benchmark III from 90nm to 45nm (to smaller node). Both figures display mean R^2 scores with the corresponding 95% confidence intervals, demonstrating INSIGHT's effectiveness across technologies. Table V summarizes the training data requirements for benchmark circuits I–V transferred from 90nm to 45nm, using a mean test R^2 score of 0.9 as the minimum acceptable performance threshold. Our experiments showed a significant $\sim$60% reduction in required training data for transfers between 90nm and 45nm, and 130nm and 90nm, while preserving high prediction fidelity. Implementing LoRA allowed us to maintain high model performance while significantly reducing computational requirements. As shown in Table VI, LoRA reduced memory footprint by $\sim$42% and training time by $\sim$25% for training size of 500 while maintaining model performance above the R^2 threshold of 0.9.

3) Inference Speed: When simultaneously processing a batch of 1,000 combinations of input parameters, INSIGHT's inference time across our benchmark circuits was measured to be $115.21\mu s$ (mean) $\pm 12.01\mu s$ (std dev), compared to several minutes for the case of commercial simulators. This result was derived from 100 independent trials per circuit on our setup.

C. Analog Sizing using INSIGHT

For these experiments, INSIGHT was pre-trained from scratch on a dataset of 1,000 randomly generated samples per topology, comprising Gaussian-sampled design parameters and their corresponding SPICE-simulated performance metrics. Note that INSIGHT does not make any a priori assumption on the underlying data distribution, and using a moderate to expert dataset would likely enhance the performance of the sizing framework. PPO-based RL agents [30] with default hyperparameters were employed, a method extensively used in EDA because of its robustness and effectiveness [6, 31], and the performance compared against two established analog circuit sizing methodologies: AutoCkt [6], a PPO-based online learning-based algorithm with default hyperparameter settings, and CRONuS [10], a state-of-the-art neural surrogate-based approach. Table VII presents the performance comparison based on the number of real-time simulations (steps) required for different circuits on 45nm technology, each row representing a test case corresponding to a fixed target specification randomly selected from the same set of designer-specified performance specifications used in [10]. The INSIGHT-based framework's performance is reported as mean $\pm$ standard deviation over 30 independent trials for robust statistical analysis. Our sizing experiments show that INSIGHT effectively reduces the critical components of Equation 2 (Section III-C) for this application, achieving $\sim$ 100-1000$\times$ improvement in computation costs over existing sizing methods on identical benchmarks and test cases, demonstrating INSIGHT's efficacy.

V. Conclusion

This paper introduces INSIGHT, a novel, technology-agnostic, and data-efficient universal neural simulator framework for analog performance prediction. By formulating the prediction task as autoregressive sequence generation, INSIGHT leverages the strengths of decoder Transformer architectures and their self-attention mechanisms to capture inherent parameter-performance interdependencies. This approach allows accurate prediction of simulation-costly critical specifications leveraging less expensive (predicted or previously available) performance metrics as conditioning information, achieving high-fidelity and data-efficient predictions across diverse circuit types and technologies. The Transformer-based architecture enables adaptive handling of variable-length design parameters and performance metrics, making INSIGHT a universal framework. INSIGHT supports compute- and data-efficient knowledge transfer across processes through parameter-efficient fine-tuning using Low-Rank Adaptation (LoRA) while maintaining high model performance. Furthermore, these fast fine-tuning capabilities make INSIGHT suitable for model-based optimization frameworks where model update time is critical. Our experiments demonstrate INSIGHT's effectiveness, consistently achieving test R^2 scores ≥ 0.95 across circuits and technologies.

To our best knowledge, this work represents the first successful application of a universal neural simulator framework using autoregressive Transformers for analog design automation. Future work will focus on developing more intelligent strategies to enhance data efficiency and extend this formulation to address more complex analog design challenges.

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